Pelton Wheel

Pelton wheel is well suited for operating under high heads. A pelton turbine has one or more nozzles discharging jets of water which strike a series of buckets mounted on the periphery of a circular disc. The *runner* consists of a circular disc with a number of buckets evenly spaced round its periphery. The *buckets* have a shape of a double semi-ellipsoidal cup. The pelton bucket is designed to deflect the jet back through 165° which is the maximum angle possible without the return jet interfering with the next bucket.

Pelton wheels are easier to fabricate and are relatively cheaper. The turbines are in general, not subjected to the cavitations effect. The turbines have access to working parts so that the maintenance or repairs can be affected in a shorter time.

Traditionally, micro hydro pelton wheels were always single jet because of the complexity and the cost of flow control governing of more than one jet.

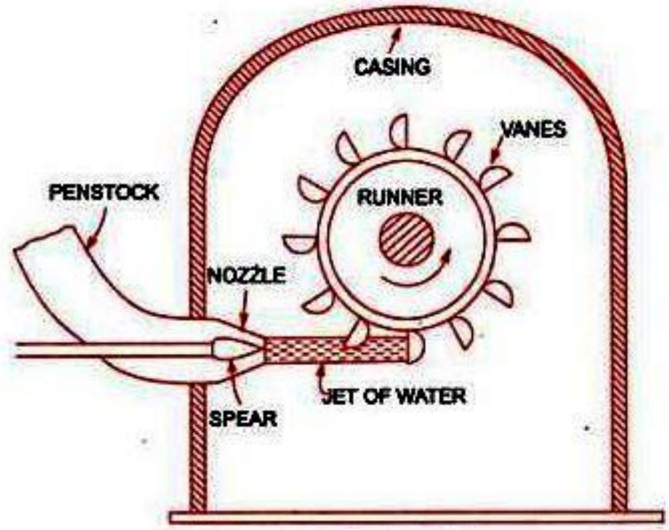
Advantages of multi-jet:

- Higher rotational speed
- Smaller runner
- Less chance of blockage

Disadvantages of multi-jet:

- Possibility of jet interference on incorrectly designed systems
- Complexity of manifolds

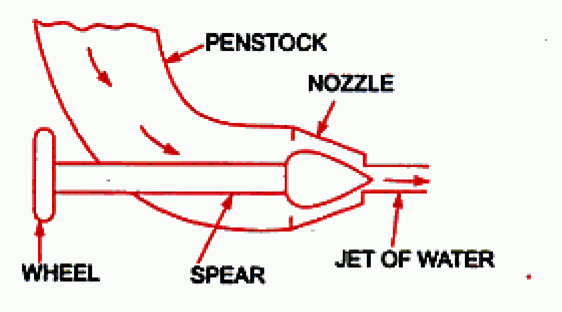
THE MAIN PARTS OF THE PELTON TURBINE ARE



Pelton turbine.

1. Nozzle and Flow regulating arrangement (spear)
2. Runner and buckets(vanes)
3. Casing
4. Breaking jet

1. Nozzle and flow Regulating Arrangement. The amount of water striking the buckets (vanes) of the runner is controlled by providing a spear in the nozzle as shown in Fig. 18.2. The spear is a conical needle which is operated either by a hand wheel or automatically in an axial direction depending upon the size of the unit. When the spear is pushed forward into the nozzle the amount of water striking the runner is reduced. On the other hand, if the spear is pushed back, the amount of water striking the runner increases.



Nozzle with a spear to regulate flow.

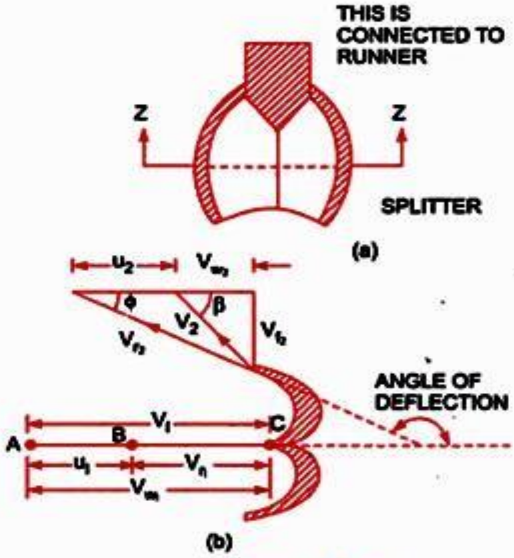
2. Runner with Buckets. Fig. 18.3 shows the runner of a Pelton wheel. It consists of a circular disc on the periphery of which a number of buckets evenly spaced are fixed. The shape of the buckets is of a double hemispherical cup or bowl. Each bucket is divided into two symmetrical parts by a dividing wall which is known as splitter.

The jet of water strikes on the splitter. The splitter divides the jet into two equal parts and the jet comes out at the outer edge of the bucket. The buckets are shaped in such a way that the jet gets deflected through 160° or 170°. The buckets are made of cast iron, cast steel bronze or stainless steel depending upon the head at the inlet of the turbine.

3. Casing. Fig. 18.4 shows a Pelton turbine with a casing. The function of the casing is to prevent the splashing of the water and to discharge water to tail race. It also acts as safeguard against accidents. It is made of cast iron or fabricated steel plates. The casing of the Pelton wheel does not perform any hydraulic function.

4. Breaking Jet. When the nozzle is completely closed by moving the spear in the forward direction, the amount of water striking the runner reduces to zero. But the runner due to inertia goes on revolving for a long time. To stop the runner in a short time, a small nozzle is provided which directs the jet of water on the back of the vanes. This jet of water is called breaking jet.

Velocity Triangles and Work done for Pelton Wheel. shows the shape of the vanes or buckets of the Pelton wheel. The jet of water from the nozzle strikes the bucket at the splitter, which splits up the jet into two parts. These parts of the jet, glides over the inner surfaces and comes out at the outer edge. Fig. 18.5 (*b*) shows the section of the bucket at *z*-*z*. The splitter is the inlet tip and outer edge of the bucket is the outlet tip of the bucket. The inlet velocity triangle is drawn at the splitter and outlet velocity triangle is drawn at the outer edge of the bucket, by the same method as explained in Chapter 17.



Shape of bucket.

Let H = Net head acting on the Pelton wheel

$= H_g - h_f$

where $\quad H_g$ = Gross head and $h_f = \dfrac{4fLV^2}{D^* \times 2g}$

where $\quad D^*$ = Dia. of Penstock, $\quad N$ = Speed of the wheel in r.p.m.

D = Diameter of the wheel, $\quad d$ = Diameter of the jet.

Then $\quad V_1$ = Velocity of jet at inlet = $\sqrt{2gH}$

$$u = u_1 = u_2 = \frac{\pi DN}{60}.$$

The velocity triangle at inlet will be a straight line where

$$V_{r_1} = V_1 - u_1 = V_1 - u$$

$$V_{w_1} = V_1$$

$$\alpha = 0 \text{ and } \theta = 0$$

From the velocity triangle at outlet, we have

$$V_{r_2} = V_{r_1} \text{ and } V_{w_2} = V_{r_2} \cos\phi - u_2.$$

The force exerted by the jet of water in the direction of motion is given by equation (17.19) as

$$F_x = \rho a V_1 [V_{w_1} + V_{w_2}] \qquad \text{...(18.8)}$$

As the angle β is an acute angle, +ve sign should be taken. Also this is the case of series of Vanes, the mass of water striking is $\rho a V_1$ and not $\rho a V_{r_1}$. In equation (18.8), 'a' is the area of the jet which is given as

$$a = \text{Area of jet} = \frac{\pi}{4} d^2.$$

Now work done by the jet on the runner per second

$$= F_x \times u = \rho a V_1 [V_{w_1} + V_{w_2}] \times u \text{ Nm/s} \qquad \text{...(18.9)}$$

Power given to the runner by the jet

$$= \frac{\rho a V_1 \left[V_{w_1} + V_{w_2}\right] \times u}{1000} \text{ kW} \qquad \text{...(18.10)}$$

Work done/s per unit weight of water striking/s

$$= \frac{\rho a V_1 \left[V_{w_1} + V_{w_2}\right] \times u}{\text{Weight of water striking/s}}$$

$$= \frac{\rho a V_1 \left[V_{w_1} + V_{w_2}\right] \times u}{\rho a V_1 \times g} = \frac{1}{g}\left[V_{w_1} + V_{w_2}\right] \times u \qquad \text{...(18.11)}$$

The energy supplied to the jet at inlet is in the form of kinetic energy and is equal to $\frac{1}{2} m V^2$

$\therefore$ K.E. of jet per second $\quad = \frac{1}{2} (\rho a V_1) \times V_1^2$

$\therefore$ **Hydraulic efficiency,** $\eta_h = \dfrac{\text{Work done per second}}{\text{K.E. of jet per second}}$

$$= \frac{\rho a V_1 \left[V_{w_1} + V_{w_2}\right] \times u}{\frac{1}{2} (\rho a V_1) \times V_1^2} = \frac{2\left[V_{w_1} + V_{w_2}\right] \times u}{V_1^2} \qquad \text{...(18.12)}$$

Now $\quad V_{w_1} = V_1,\ V_{r_1} = V_1 - u_1 = (V_1 - u)$

$\therefore \quad V_{r_2} = (V_1 - u)$

and $\quad V_{w_2} = V_{r_2} \cos\phi - u_2 = V_{r_2} \cos\phi - u = (V_1 - u)\cos\phi - u$

Substituting the values of V_{w_1} and V_{w_2} in equation (18.12),

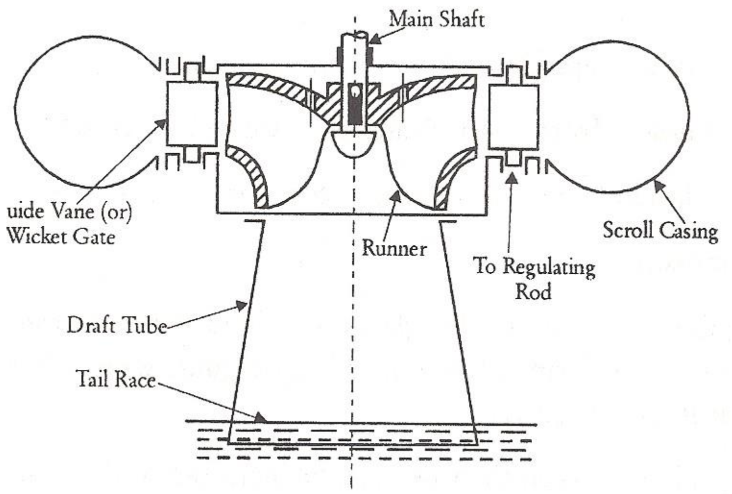
$$\eta_h = \frac{2\left[V_1 + (V_1 - u)\cos\phi - u\right] \times u}{V_1^2}$$

$$= \frac{2\left[V_1 - u + (V_1 - u)\cos\phi\right] \times u}{V_1^2} = \frac{2(V_1 - u)[1 + \cos\phi]u}{V_1^2}. \qquad \text{...(18.13)}$$

Maximum Efficiency = ——— ....u =—

Francis Turbine

Francis turbine is a mixed flow type, in which water enters the runner radially at its outer periphery and leaves axially at its centre. Fig.5.6 illustrates the Francis turbine. The runner blades are profiled in a complex manner and the casing is scrolled to distribute water around the entire perimeter of the runner. The water from the penstock enters a *scroll case* which completely surrounds the runner. The purpose of the scroll case is to provide an even distribution of water around the circumference of the turbine runner, maintaining an approximately constant velocity for the water so distributed. The function of *guide vane* is to regulate the quantity of water supplied to the runner and to direct water on to the runner at an angle appropriate design. A *draft tube* is a pipe or passage of gradually increasing cross sectional area which connects the runner exit to the tail race.



Description of main parts

Francis turbine consists mainly of the following parts

a) Spiral or scroll casing
b) Guide mechanism
c) Runner and turbine main shaft
d) Draft tube

Spiral casing of scroll casing

The casing of the francis turbine is designed in a spiral form with a gradually increasing area. The advantages of this design are

i) Smooth and even distribution of water around the runner.
ii) Loss of head due to the formation of eddies is avoided.
iii) Efficiency of flow of water to the turbine is increased.

- In big units stay vanes are provided which direct and water to the guide vanes.
- The casing is also provided with inspection holes and pressure gauge connection.
- The selection of material for the casing depends upon the head of water to be supplied

For a head – up to 30 metres – concrete is used.

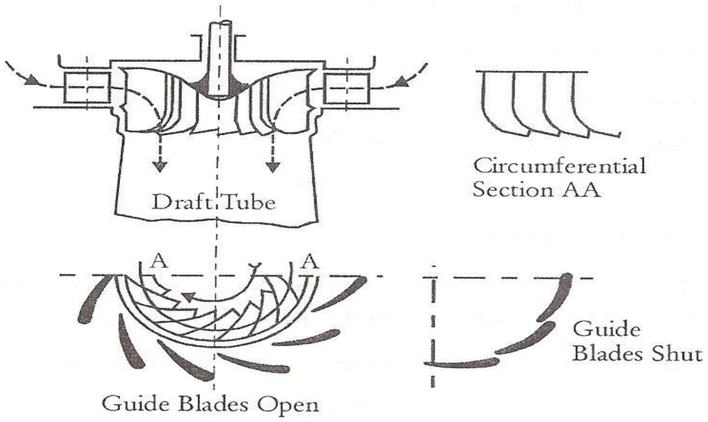
For a head – from 30 to 60 metres – welded rolled steel plates are used.

For a head of above 90 metres – cast steel is used.

Guide mechanism

The guide vanes of wicket gates are fixed between two rings. This arrangement is in the form of a wheel and called guide wheel. Each vane can be rotated about its pivot centre.

The opening between the vanes can be increased or decreased by adjusting the guide wheel. The guide wheel is adjusted by the regulating shaft which is operated by a governor.



The guide mechanism provides the required quantity of water to the runner depending upon the load conditions. The guide vanes are in general made of cast steel.

Runner and turbine main shaft

The flow in the runner of a modern Francis turbine is partly radial and partly axial. The runners may be classified as

i) Slow ii) Medium iii) Fast

The runner may be cast in one piece or made of separate steel plates welded together. The runners are made of CI for small output, cast steel or stainless steel or bronze for large output. The runner blades should be carefully finished with high degree of accuracy.

The runner may be keyed to the shaft which may be vertical or horizontal. The shaft is made of steel and is forged it is provided with a collar for transmitting the axial thrust.

Draft tube

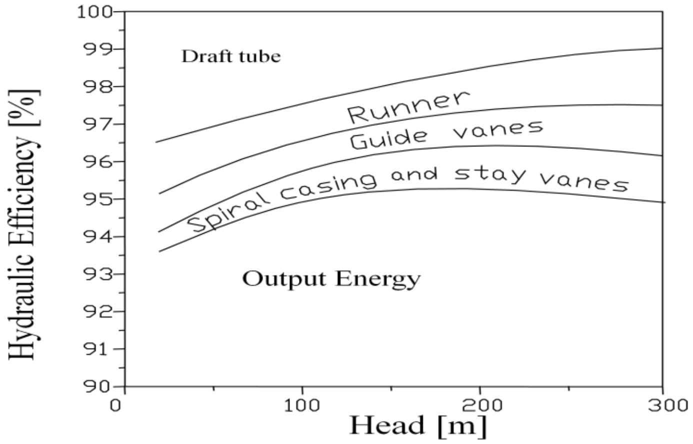
The water after doing work on the runner passes on to the tail race through a tube called draft tube. It is made of riveted steel plate or pipe or a concrete tunner. The cross – section of the tube increases gradually towards the outlet. The draft tube connects the runner exit to the tail race. This tube should be drowned approximately 1 metre below the tail race water level.

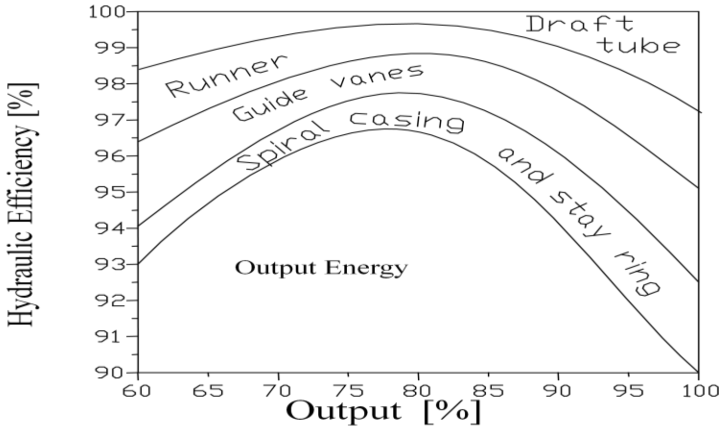
Working Principles of Francis Turbine

The water is admitted to the runner through guide vanes or wicket gates. The opening between the vanes can be adjusted to vary the quantity of water admitted to the turbine. This is done to suit the load conditions.

The water enters the runner with a low velocity but with a considerable pressure. As the water flows over the vanes the pressure head is gradually converted into velocity head. This kinetic energy is utilized in rotating the wheel. Thus the hydraulic energy is converted into mechanical energy. The outgoing water enters the tail race after passing through the draft tube. The draft tube enlarges gradually and the enlarged end is submerged deeply in the tail race water. Due to this arrangement a suction head is created at the exit of the runner.

Losses in Francis Turbines





VELOCITY TRIANGLE

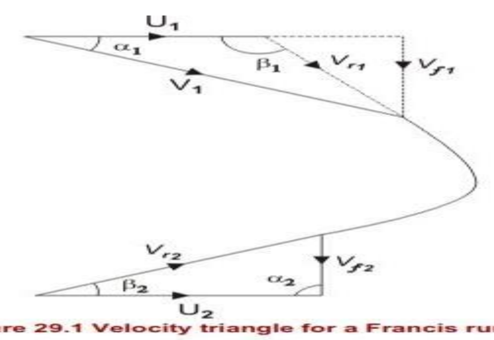
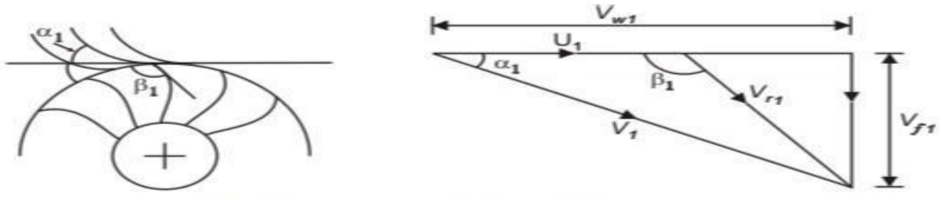


Figure 29.1 Velocity triangle for a Francis runner

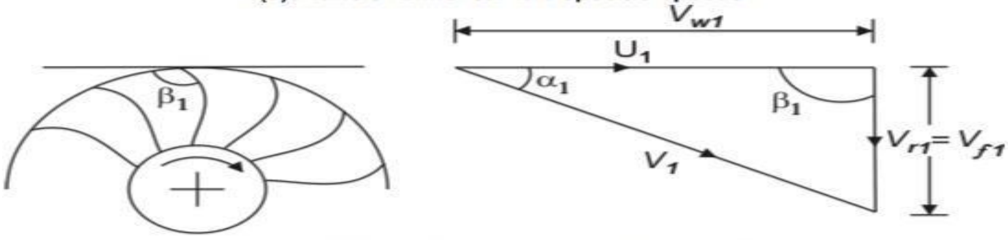
Work done per second on the runner per second will be = $V_{w1}U_1]$

Work done per second per unit weight of water striking/s = $[V_{w1}U_1]$

Hydraulic Efficiency will be given by = ______



(a) Francis runner for low specific speeds



(b) Francis runner for normal specific speeds

Kaplan Turbine

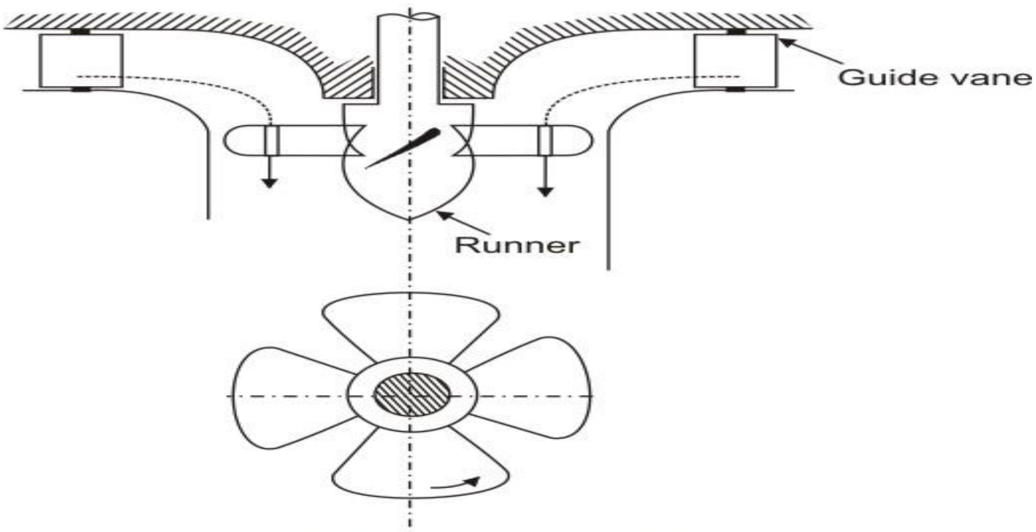


Fig. 30.3 A propeller of Kaplan turbine

It is an axial flow turbine which is suitable for relatively low heads. It will be seen that the main components of Kaplan turbine such as scroll casing, guide vanes, and the draft tube are similar to those of a Francis turbine.

This type of turbine evolved from the need to generate power from much lower pressure heads than are normally employed with the Francis turbine. To satisfy large power demands very large volume flow rates need to be accommodated in the Kaplan turbine, i.e. the product QHE is large. The overall flow configuration is from radial to axial. Figure 9.16 is a part sectional view of a Kaplan turbine in which the flow enters from a volute into the inlet guide vanes which impart a degree of swirl to the flow determined by the needs of the runner. The flow leaving the guide vanes is forced by the shape of the passage into an axial direction and the swirl becomes essentially a free vortex.

The vanes of the runner are similar to those of an axial-flow turbine rotor but designed with a twist suitable for the free-vortex flow at entry and an axial flow at outlet. Because of the very high torque that must be transmitted and the large length of the blades, strength considerations impose the need for large blade chords.

As a result, pitch/chord ratios of 1.0 to 1.5 are commonly used by manufacturers and, consequently, the number of blades is small, usually 4, 5 or 6. The Kaplan turbine incorporates one essential feature not found in other turbine rotors and that is the setting of the stagger angle can be controlled.

At part load operation the setting angle of the runner vanes is adjusted automatically by a servo mechanism to maintain optimum efficiency conditions. This adjustment requires a complementary adjustment of the inlet guide vane stagger angle in order to maintain an absolute axial flow at exit from the runner.

Basic equations

Most of the equations presented for the Francis turbine also apply to the Kaplan (or propeller) turbine, apart from the treatment of the runner. Figure 9.17 shows the velocity triangles and part section of a Kaplan turbine drawn for the mid-blade height. At exit from the runner the flow is shown leaving the runner without a whirl velocity, i.e. c 3 D 0 and constant axial velocity.

The theory of free-vortex flows was expounded in Chapter 6 and the main results as they apply to an incompressible fluid are given here. The runner blades will have a fairly high degree of twist, the amount depending upon the strength of the circulation function K and the magnitude of the axial velocity. Just upstream of the runner the flow is assumed to be a free-vortex and the velocity components are accordingly:

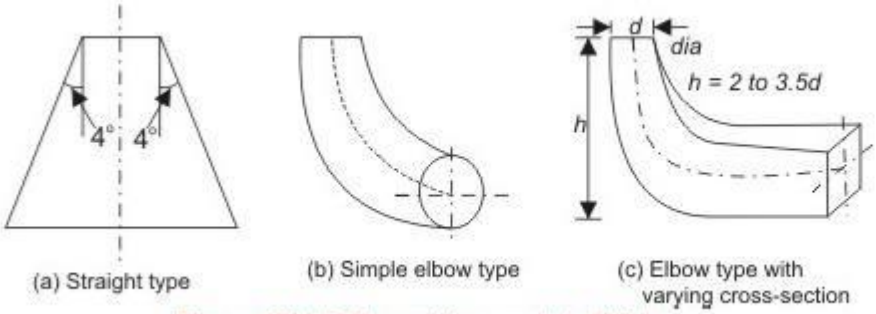
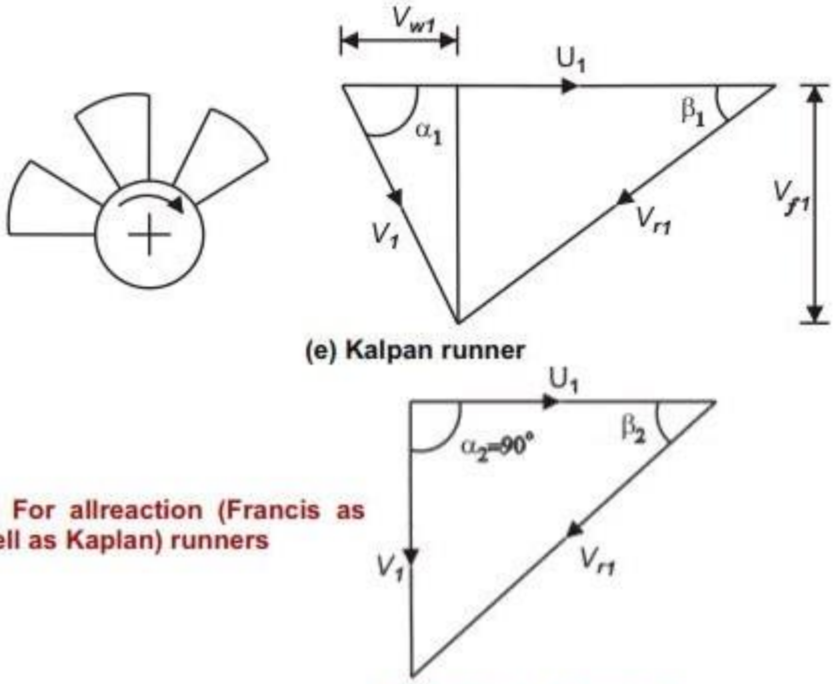


Figure 30.4 Different types of draft tubes

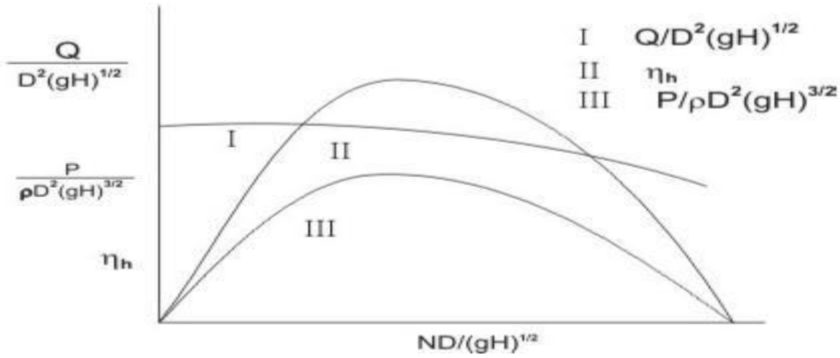


Outlet velocity triangle

Fig. 30.2 Evolution of Kaplan runner form Francis one

Performance Characteristics of Reaction Turbine

It is not always possible in practice, although desirable, to run a machine at its maximum efficiency due to changes in operating parameters. Therefore, it becomes important to know the performance of the machine under conditions for which the efficiency is less than the maximum. It is more useful to plot the basic dimensionless performance parameters is derived earlier from the similarity principles of fluid machines. Thus one set of curves, as shown in Fig. 31.1, is applicable not just to the conditions of the test, but to any machine in the same homologous series under any altered conditions.



Performance characteristics of a reaction turbine (in dimensionless parameters)

Below figure is one of the typical plots where variation in efficiency of different reaction turbines with the rated power is shown.

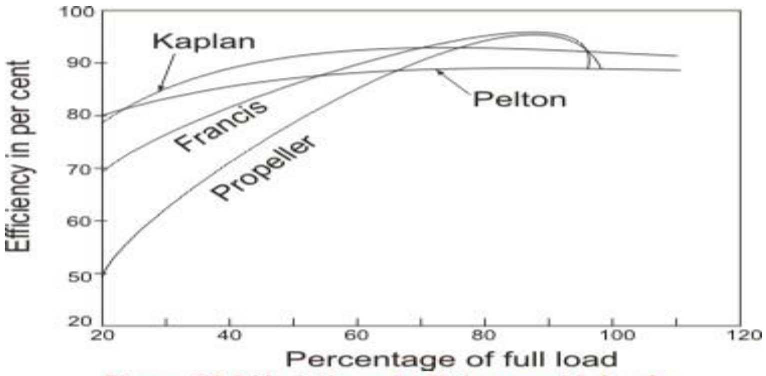


Figure 31.2 Variation of efficiency with load

Characteristic Curves

The turbines are generally designed to work at particular values of H, Q, P, N and $_o$ which are known as the designed conditions. It is essential to determine exact behaviour of the turbines under the varying conditions by carrying out tests either on the actual turbines or on their small scale models. The results of these tests are usually graphically represented and the resulting curves are known as characteristic curves.

- -constant head characteristic curves
- -constant speed characteristic curves
- -constant efficiency characteristic curves

In order to obtain constant head characteristics curves the tests are performed on the turbine by maintaining a constant head and a constant gate opening and the speed is varied by changing the load on the turbine. A series of values of N are thus obtained and corresponding to each value of N, discharge Q and the output power P are measured. A series of such tests are performed by varying the gate opening, the head being maintained constant at the previous value. From the data of the tests the values of Qu, Pu, nu and $_o$ are computed for each gate opening. Then with Nu as abscissa the values of Qu, Pu and $_o$ for each gate opening are plotted. The curves thus obtained for pelton wheel and the reaction turbines for four different gate openings are shown in Fig.

Governing Mechanism of Turbines

Usually hydraulic turbines are coupled to the generators. The power output of the generator is actually the load on the turbine. When the load changes the speed or the turbine also change.

The governing of a turbine is the operation by which the speed of the turbine is kept constant by controlling the discharge irrespective of the fluctuations of load on the turbine.

Functions of governor

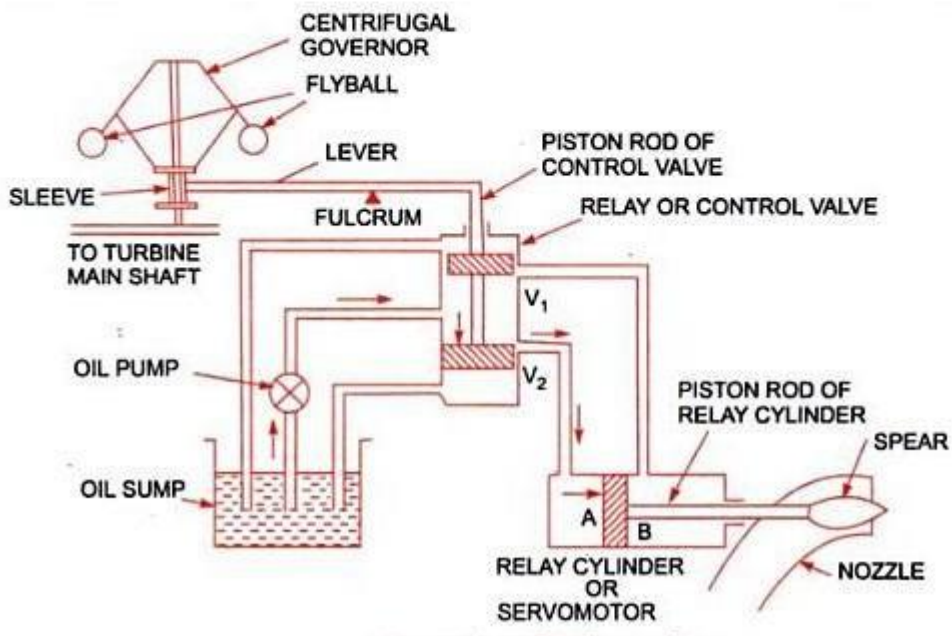
The main functions of a governor are

i) Noticing the speed variations quickly.
ii) Operating the different components rapidly and effectively.
iii) Controlling the discharge to maintain constant speed.
iv) Matching the speed of the turbine and generator.
v) Setting the amount of load to the turbine unit.
vi) To alert the operator under extreme conditions.

Working:

Fig Shows governing mechanism of a pelton wheel. It consists of a spring loaded centrifugal or pendulum. This governor is driven by the turbine main shaft by a belt.

As the speed increases, the balls of the governor moves outward and as speed falls the balls move inwards. The pendulum operates a small control valve and the spear is operated by a servomotor.



Governing of Pelton turbine.

If the load decreases the speed of the turbine will increases. As the speed increases, the pendulum starts moving faster. This causes an upward movement of the sleeve attached to the governor. As the control valve is moved in the upward, direction, high pressure oil is admitted to the left hand side of the servomotor piston. This causes the spear to move deep into the nozzle hole reducing the discharge of water directed towards the vanes. Now the oil from the right hand end of the servomotor cylinder is returned to the oil sump.

When the load on the turbine increases, the main level moves upward. This pulls the piston of the control valve upward. Now oil under pressure enters the right hand side of the servo – motor piston. The piston moves to the left and the spear also moves to the left and there is an increase in flow through nozzle. The increased discharge increases the output and the normal speed is maintained.

Draft Tube:

- The draft tube is pipe of gradually increasing area which connects the outlet of the runner to the tail race.
- It is used for discharging water from the exit of the turbine to the tail race.
- This pipe of gradually increasing area is called draft tube.
- One end of the draft tube is connected to the outlet of the runner while the other end is sub-merged below the level of water in the tail race.

Functions of Draft Tube:

- It permits a negative head to be established at the outlet of the runner and thereby increase the net head on the turbine.
- The turbine may be placed above the tail race without any loss of net head and hence turbine may be inspected properly.
- It converts a large proportion of the kinetic energy ($V_2^2/ 2g$) rejected at the outlet of the turbine into useful pressure energy.
- Without the draft tube the kinetic energy rejected at the outlet of the turbine will go waste to the tail race.
- Hence by using draft tube net head on the turbine increases.
- The turbine develops more power and also the efficiency of the turbine increases.
- If the reaction turbine is not fitted with a draft tube, the pressure at the outlet is equal to the atmospheric pressure.

Types of Draft Tube:

- Conical Draft Tubes
- Simple elbow Tubes
- Elbow Draft tubes with circular inlet and rectangular outlet.
- Moody's bell mouthed tube

Draft Tube Theory:

Consider a capital draft-tube as shown in figure

H_s = vertical height of draft tube above the tail race.

Y= distance of bottom of draft tube from tail race.

Applying Bernoulli's equation to inlet (section 1-1) and outlet (section 2-2) of the draft tube and taking section 2-2 as datum line, we get

$$— + — + (H_s + y) = — + — + 0 + h_f \text{ ----------------- } (1)$$

Where h_f = loss of energy between sections 1-1 and 2-2.

But $— =$ Atmospheric pressure head + y

$$= \cancel{+\ y}$$

Substituting this value in equation (1), we get

$$— + — + (H_s + y) = — + y + — + h_f$$

$$\boxed{— \quad \underline{=} - H_s - - — \quad — -}$$

Efficiency of Draft Tube:

The efficiency of draft tube is defined as the ratio of actual conversation of kinetic head into pressure head in the draft tube to the kinetic head at the inlet of the draft tube.

$$_d =$$

1. A water turbine has a velocity of 6 m/s at the entrance to the draft tube and a velocity of 1.2 m/s at the exit. For friction losses of 0.1 m and tail water 5 m below the entrance to the draft-tube, find the pressure head at the entrance.

   Given:

   Velocity at inlet (V_1) = 6 m/s
   Velocity at outlet (V_2) = 1.2 m/s
   Friction loss (h_f) = 0.1 m
   H_s = 5 m

   To find:

   Pressure Head at entrance =?

   Solution:

   Atmospheric pressure (P_a = 1.0135 bar = 1.0135 x 10^5 N/m^2)

   $$= 10.3 \text{ m}$$

   $$_s -$$

   $$= \mathbf{-\ 6.6616\ m}$$

2. A conical draft tube having diameter at the top as 2.0 m and pressure head at 7m of water(vacuum), discharge water at the outlet with a velocity 0f 1.2 m/s at the rate of 25 m^3/s. If atmospheric pressure head is 10.3 m of water and losses between the inlet and outlet of the draft-tubes are negligible, find the length of draft-tube immersed in water. Total length of tube is 5 m.

   Given:

   Diameter at top, D_1 = 2 m
   Pressure head = -7m (vacuum) = 10.3 – 7 = 3.3 m (abs)
   Velocity at outlet, V_2 =1.2 m/s
   Q = 25 m^3/s
   H_f = negligible
   Total length of the tube = 5 m

   $$= 10.3 \text{ m}$$

   To find:

   Length of the tube immersed in water (y) =?

   Solution:

   $Q = A_1V_1$
   V_1 = 7.957 m/s

   $$_s -$$

   $$\mathbf{H_s = 3.8464m}$$

   Y = total length – H_s = 5 – 3.8464 = **1.1536m**

3. A conical draft tube having inlet and outlet diameter 1 m and 1.5 m discharges water at outlet with a velocity of 2.5 m/s. The total length of the draft tube is 6 m and 1.2 m of the length of the draft tube is immersed in water. If the atmospheric pressure head is 10.3 m of water and loss of head due to friction in the draft tube is equal to 0.2 x velocity head at outlet of the tube, find: (i) pressure head at inlet,(ii) efficiency of the draft tube.

Given:

Diameter at inlet (D_1) = 1.0 m

Diameter at outlet (D_2) = 1.5 m

Velocity at outlet (V_2) = 2.5 m/s

Total length of tube, $H_s + y = 6$ m

$y = 1.2$ m

$H_s = 4.80$ m

— = 10.3 m

Loss of head due to friction (h_f) = 0.2 x velocity of head at outlet

= 0.2 x — 0.0637 m

To find:

Pressure Head at entrance =?

d =?

Solution:

$Q = A_2V_2$

$A_2 = \underline{x}\ D^2_2$

$Q = 4.4178\ m^3/s$

$Q = A_1V_1$

$A_1 = \underline{x}\ D^2_1$

$V_1 = 5.625$ m/s

(i) Pressure head at inlet

— _ — s - — - — -

— = **4.27 m (abs)**

(ii) Efficiency of draft tube

d = —- — - / —

d = **76.3 %**

Pelton Wheel

Formulas:

- Work done by the jet per second = $aV_1 [V_{w1} \pm V_{w2}] X u$
- Runner power =
- Water power =
- Kinetic energy = V ^{3} 1
- Hydraulic Efficiency = =
- Velocity of jet at inlet is given by $V_1 = C_v$, H =

  C_V = Co-efficient of velocity = 0.98 or 0.99

  - Gross head $H_g = h_f + H$
  - Gross Head $H_g = h_f + H$ + head lost in nozzle------- (if nozzle loss consider)
  - Head lost in nozzle = (– 1)
  - Efficiency of nozzle = $(H_g – h_f)/ H_g$
- Velocity of wheel is given by (u) = C_u ,

  C_u = speed ratio = 0.43 to 0.48

  - $u = u_1 = u_2$ =
- **Jet ratio(m)** = D/d

  D = pitch diameter, d = jet diameter
- Number of buckets(Z) = 15 +
- No of jet =

  Q = total discharge, q = discharge by single jet
- Mechanical Efficiency = m =
- Volumetric Efficiency = v =
- Overall efficiency = o =

  o = m x h

  - $V_{r1} = V_{r2}$

1. A pelton wheel has a mean bucket speed of 10 meters per second with a jet of water flowing at the rate 700 litres/s under a head of 30 meters. The buckets deflect jet through an angle of 160°. Calculate the power given by water to the runner and the hydraulic efficiency of the turbine. Assume co-efficient of velocity as 0.98.

Given:

$u = u_1 = u_2 = 10$ m/s

Q = 700 litres/s = 0.7

m^3/s H = 30 m

=180° – 160° = 20°

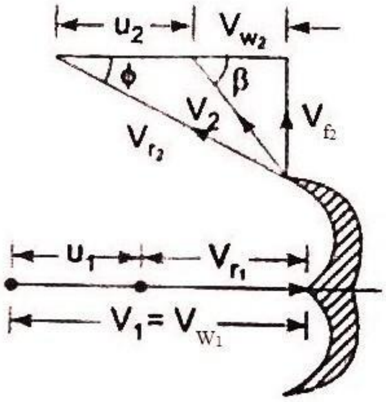
$C_V = 0.98$

To find:

Runner power =?

$_h$ =?

Solution:



$V_1 = C_V$ = 23.77 m/s

From inlet velocity triangle

$V_{r1} = V_1 - u_1 = 13.77$ m/s

$V_{w1} = V_1 = 23.77$ m/s

From outlet velocity triangle

$V_{r2} = V_{r1} = 13.77$ m/s

=

$V_{w2} = 2.94$ m/s

a. Runner Power

Runner power =

$Q = aV_1 = 0.7\ m^3/s$

R.P = 186.97 kW

b. Hydraulic Efficiency

$_h$ =

$_h$ = 94.54 %

2. A Pelton wheel is to be designed for the following specification: shaft power = 11,772; Head = 380 meters; Speed = 750 R.p.m; Overall efficiency = 86 %; Jet diameter is not to exceed one-sixth of the wheel diameter. Determine: (i) the wheel diameter, (ii) the number of jets required, and (iii) diameter of the jet. Take $C_{v1} = 0.985$ and $C_{u1} = 0.45$.

Given:

S.P = 11,772 kW

H = 380 m

N =750 r.p.m

$_o$ = 0.86

d = 1/6 D

$C_V = 0.985$

$C_u = 0.45$

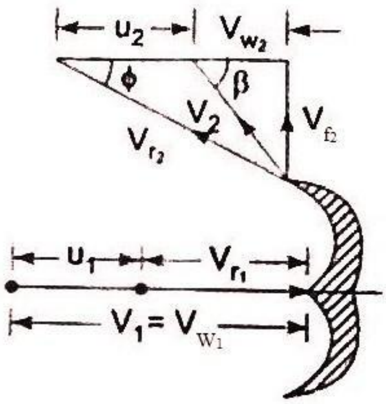
To find:

Wheel Diameter (D) =?

Jet Diameter (d) =?

No of Jets =?

Solution:



$V_1 = C_V = 85.05$ m/s

$u = u_1 = u_2 = C_u = 38.85$ m/s

a. The wheel Diameter

$u = u_1 = u_2 =$

D = 0.989 m

b. Jet Diameter

d = 1/6 D

d= 0.165 m

c. No of jets

q = Area of jet x velocity of jet = 1.818 m^3/s

$_o = 0.86 =$

W.P = 13688 kW

Water power = , where,

W = g = 1000 x 9.81

Q = 3.672 m^3/s

No of jet = **= 2 jets**

3. The penstocks supply water from a reservoir to the pelton wheel with a gross head of 500m. One third of the gross head is lost in friction in the penstock. The rate of flow of water through the nozzle fitted at the end of the penstock is 2.0m^3/s. The angle of deflection of the jet is 165°. Determine the power given by the water to the runner and also hydraulic efficiency of the pelton wheel. Take speed ratio = 0.45 and C_V = 1.0.

Given:

$H_g = 500$ m

$h_f =$ = 166.7 m

Net head $H = H_g – h_f = 333.30$ m

Q = 2.0 m^3/s

=180° – 165° = 15°

Speed ration C_u =0.45

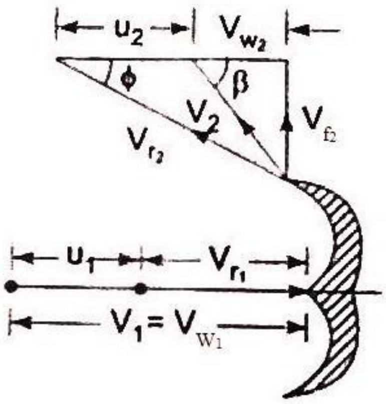
Co-efficient of velocity C_V = 1.0

To find:

Runner power =?

$_h$ =?

Solution:



$V_1 = C_V = 80.86$ m/s

$u = u_1 = u_2 = C_u = 36.387$ m/s

From inlet velocity triangle

$V_{r1} = V_1 – u_1 = 44.473$ m/s

$V_{w1} = V_1 = 80.86$ m/s

From outlet velocity triangle

$V_{r2} = V_{r1} = 44.473$ m/s

=

$V_{w2} = 6.57$ m/s

a. Runner Power

Runner power =

Q = aV_1 = 2.0 m^3/s

R.P = 6362.63 kW

b. Hydraulic Efficiency

$_h =$

$_h$ = 97.31 %

4. A Pelton wheel is having a mean bucket diameter of 1 m and is running at 1000 r.p.m. The net head on the Pelton wheel is 700 m. If the side clearance angle is 15° and discharge through nozzle is $0.1 m^3/s$, find: (i) power available at the nozzle, and (ii) Hydraulic efficiency of the turbine.

Given:

D = 1.0 m

N = 1000 r.p.m

H = 700 m

= 15°

$Q = 0.1 m^3/s$

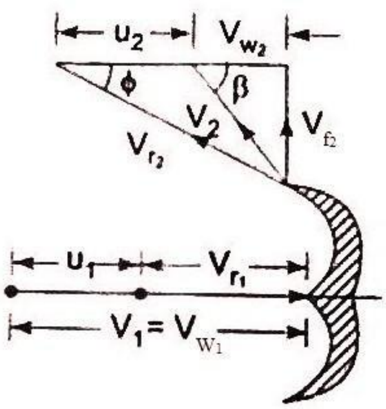
C_V = 1 ------ (not given)

To find:

Power available at nozzle =?

$_h$ =?

Solution:



$V_1 = C_V$ = 117.19 m/s

$u = u_1 = u_2 =$ —— = 52.36 m/s

From inlet velocity triangle

$V_{r1} = V_1 - u_1$ = 64.83 m/s

$V_{w1} = V_1$ = 117.19 m/s

From outlet velocity triangle

$V_{r2} = V_{r1}$ = 64.83 m/s

=

V_{w2} = 10.26 m/s

a. **Power available at nozzle**

Water power = ———, where,

W = g = 1000 x 9.81

W.P = **686.7 kW**

b. **Hydraulic Efficiency**

$_h$ =

$_h$ = 98.18 %

5. A Pelton wheel is working under a gross head of 400 m. The water is supplied through penstock of diameter 1 m and length 4 km from reservoir to the Pelton wheel. The co-efficient of friction for the penstock is given as 0.008. The jet of water of diameter 150mm strikes the buckets of the wheel and gets deflected through an angle of 165°. The relative velocity of the water at outlet is reduced by 15% due to friction between inside surface of the bucket and water. If the velocity of the buckets is 0.45 times the jet velocity at inlet and mechanical efficiency as 85 % determine: (i) power given to the runner, (ii) shaft power (iii) hydraulic efficiency and overall efficiency.

Given:

H_g = 400 m
D = 1.0 m
L = 4000 m
f = 0.008
d = 0.15 m
=180° – 165° = 15°
$V_{r2} = 0.85 V_{r1}$
u = 0.45 x jet velocity
$_m$ = 0.85

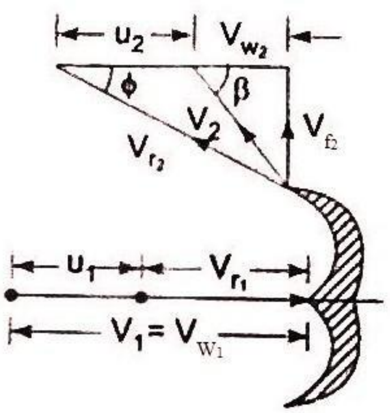
To find:

Runner power =?

Shaft power =?

$_h$ =?

Solution:



Let, V^* = velocity of penstock

V_1 = Velocity of jet of water

Using continuity equation

Area of penstock x V^* = Area of jet x V_1

— x D^2 x V^* = x d^2 x V_1

$V^* = 0.0225V_1$---------(1)

Gross head H_g =h_f + H

400 = —— + ------- (2)

Sub (1) in (2), we get

V_1 = 85.83 m/s

u = 0.45 x jet velocity = 38.62 m/s

From inlet velocity triangle

$V_{r1} = V_1 – u_1 = 47.21$ m/s

$V_{w1} = V_1 = 85.83$ m/s

From outlet velocity triangle

$V_{r2} = 0.85V_{r1} = 40.13$ m/s

=

$V_{w2} = 0.143$ m/s

a = $\cancel{x}$ $d^2 = 0.01767\ m^2$

a. **Runner Power**

Runner power =

R.P = 5033.54 kW

b. **Shaft Power**

m —

Shaft power = 4278.5 kW

c. **Hydraulic Efficiency**

h =

h = 90.14 %

6. A Pelton wheel nozzle, for which C_V = 0.97, is 400m below the water surface of a lake. The jet diameter is 80 mm, the pipe diameter is 0.6 m, its length is 4 km and f = 0.032 in the formula **$h_f = fLV^2/ 2g \times d$**. The buckets, deflects the jet through 165° and they run at 0.48 times the jet speed, bucket friction reducing the relative velocity at outlet by 15% of relative velocity at inlet. Mechanical efficiency = 90%. Find the flow rate and the shaft power developed by the turbine.

Given:

$C_V = 0.97$

$H_g = 400$ m

d = 0.08 m

D = 0.6 m

L = 4000 m

f = 0.032

=180° – 165° = 15°

u = 0.48 x jet speed

$V_{r2} = 0.85V_{r1}$

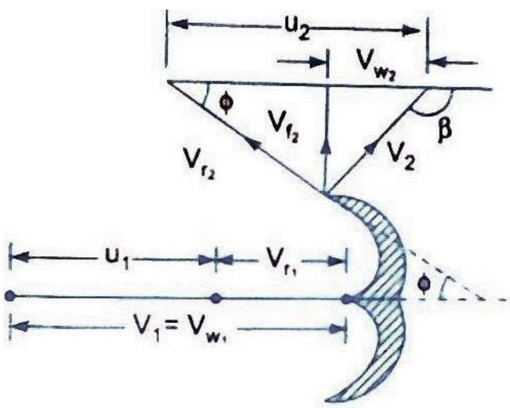
m = 0.90

To find:

Flow rate (Q) =?

Shaft power =?

Solution:



Let, V^* = velocity of penstock

V_1 = Velocity of jet of water

Using continuity equation

Area of penstock x V^* = Area of jet x V_1

_ x D^2 x V^* = x d^2 x V_1

$V^* = 0.0177V_1$ -------- (1)

Gross Head **(H_g) = h_f + H + head lost in nozzle**

400 = + + — — — (— – 1) --------- (2)

Sub (1) in (2), we get

V_1 = 83.47 m/s

u = 0.48 x jet velocity = 40.06 m/s

From inlet velocity triangle

$V_{r1} = V_1 - u_1 = 43.41$ m/s

$V_{w1} = V_1 = 83.47$ m/s

From outlet velocity triangle

$V_{r2} = 0.85V_{r1} = 36.898$ m/s

=

$V_{w2} = -4.42$ m/s

a = x d^2 = 0.0050265 m^2

a. **Flow rate**

Q =a V_1 = 0.419 m^3/s

b. **Shaft Power:**

m

Runner power =

R.P = 1326.865 kW

S.P = 1326.865 x 0.90 = 1194.18 kW

7. A 137 mm diameter jet of water issuing from a nozzle impinges on the buckets. The head available at the nozzle is 400m. Assuming $C_V = 0.97$, speed ratio as 0.46, and reduction in relative velocity while passing through buckets as 15%, find: (i) the force exerted by the jet on buckets in tangential direction, (ii) the power developed.

Given:

d = 0.137 m

=180° – 165° = 15°

H = 400 m

$C_V = 0.97$

$C_u = 0.46$

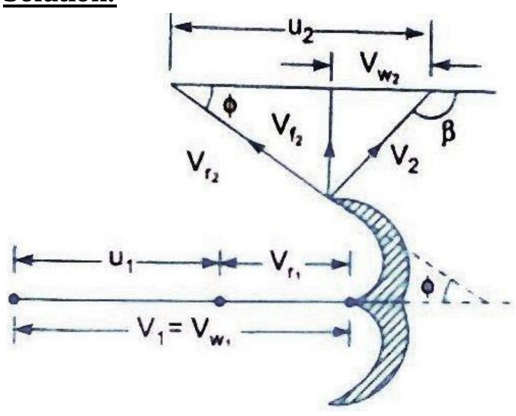
$V_{r2} = 0.85 V_{r1}$

To find:

Force Exerted =?

Runner power =?

Solution:



$V_1 = C_V = 85.93$ m/s

$u = u_1 = u_2 = C_u = 40.75$ m/s

From inlet velocity triangle

$V_{r1} = V_1 - u_1 = 45.18$ m/s

$V_{w1} = V_1 = 85.93$ m/s

From outlet velocity triangle

$V_{r2} = 0.85V_{r1} = 38.40$ m/s

=

$V_{w2} = -3.658$ m/s

a. **Force exerted**

F_X =

= **104206 N**

b. **Runner Power**

Runner power =

R.P = 4246.4 kW

8. Two jets strike the buckets of a Pelton wheel, which is having shaft power at 15450 kW. The diameter of each jet is given as 200 mm. If the net head on the turbine is 400 m. find the overall efficiency of the turbine. Take C_V = 1.0.

Given:

No of jets = 2

S.P = 15450 kW

d = 0.20 m

H = 400 m

C_V = 1.0

To Find:

Overall efficiency = $_o$ =?

Solution:

$V_1 = C_V = 88.58$ m/s

Discharge of each jet (q) = a x V_1 = 2.78 m^3/s

Total Discharge (Q) = 2 x q = 5.56 m^3/s

Water power = ~~, where~~,

W = g = 1000 x 9.81

W.P = 21817.44 **kW**

$_o$ = **70.8 %**

9. The water available for a Pelton wheel is 4 cumec and the total head from the reservoir to the nozzle is 250m. The turbine has two runners with two jets per runner. All the four jets have the same diameters. The pipe line is 3000 m long. The efficiency of power transmission through the pipe line and the nozzle is 91% and efficiency of each runner is 90%. The velocity co-efficient of each nozzle is 0.975 and co-efficient of friction '4f' for the pipe is 0.0045. Determine: (i) The power developed by the turbine (ii) The diameter of the jet, and (iii) The diameter of the pipe line.

Given:

Q = a V_1 = 4 cumec = 4 m^3/s

H_g = 250 m

No of jets = 2 x 2 = 4

L =3000 m

Efficiency of pipeline or nozzle = 0.91

Efficiency of runner or $_h$ = 0.90

C_V= 0.975

4f = 0.0045

To Find:

Power developed =?

D =?

d =?

Solution:

Efficiency of nozzle = $(H_g - h_f)/ H_g$

h_f = 22.5 m

Net Head H = $H_g - h_f$ = 227.5 m

$V_1 = C_V = 65.14$ m/s

a. **Power developed**

Hydraulic Efficiency =

Kinetic energy =

= 8486.44 x 10^3 Nm

W.D by the jet = 0.90 x K.E = 7637.8 x 10^3 Nm

Power developed = W.D/ 1000 = **7637 kW**

b. **Diameter of the jet**

No of jets = Q/q

q = 1.0 m^3/s

q= ~~x~~ d^2 x V_1

d = 0.14 m

c. **Diameter of pipe line**

Q = A x V^*

4 = ~~x~~ D^2 x V^*

$V^* = 5.09/D^2$ --------- (1)

h_f =

22.5 =

22.5 =

D^5 = 0.7933

D = $(0.7933)^{1/5}$ = **0.995 m**

10. The following data is related to Pelton wheel: Head at the base of the nozzle = 80 m; Diameter of the jet = 100 mm; Discharge of the nozzle – 0.30 m^3/s; Power at the shaft = 206 kW; Power absorbed in mechanical resistance = 4.5 kW; Determine (i) Power lost in nozzle and (ii) Power lost due to hydraulic resistance in the runner.

Given:

H = 80 m

d = 0.1 m

Q = 0.30 m^3/s

S.P = 206 kW

Power absorbed in mechanical resistance = 5.5 kW

To Find:

Power lost in nozzle =?

Power lost due to hydraulic resistance =?

Solution:

Q =a V_1

V_1 = 38.197 m/s

Power at the nozzle

Water power = ~~, where~~,

W = g = 1000 x 9.81

W.P **= 235.44 kW**

Power at the jet = K.E/ 1000

Kinetic energy = – –

Power at the jet **= 218.85 kW**

a. **Power lost in nozzle**

Power at the Nozzle = Power at the jet + Power lost in nozzle

Power lost in nozzle = **16.59 kW**

b. **Power lost due to hydraulic resistance:**

Power at the jet = Power at the shaft + Power absorbed in mechanical resistance + Power lost in hydraulic resistance

Power lost in hydraulic resistance = **8.35 kW**

Design of Pelton Wheel:

Design of Pelton wheel means the following data is to be determined

- Diameter of the jet(d)
- Diameter of the wheel(D)
- Width of the buckets which is = 5 x d
- Depth of the buckets which is = 1.2 x d
- Number of buckets on the wheel

Size of the buckets means the width and the depth of the buckets

11. A Pelton wheel is to be designed for a head of 60m when running at 200 r.p.m. The Pelton wheel develops 95.6475 kW shaft Power. The velocity of the buckets = 0.45 times the velocity of the jet, overall efficiency = 0.85 and co-efficient of the velocity is equal to 0.98.

Given

H =60 m

N = 200 r.p.m

S.P = 95.6475 kW

u = 0.45 x jet velocity

$_o$ = 0.85

C_V = 0.98

To Find

Diameter of the jet (d) =?
Diameter of the wheel (D) =?
Width of the buckets which is =?
Depth of the buckets which is =?
Number of buckets on the wheel =?

Solution:

a. **Diameter of jet (d) :**

$_o$ — = 0.85

W.P = 81.3 kW

Water power = , where,

W = g = 1000 x 9.81

$Q = 0.1912\ m^3/s$

$V_1 = C_V = 33.62$ m/s

$Q = A\ V_1$

$Q = x\ d^2 \times V_1$

d = 0.085 m

b. **Diameter of wheel**:

$u_1 = 0.45 \times V_1 = 15.13$ m/s

$u = u_1 = u_2 =$

D = 1.44 m

c. **Width of the buckets:**

Width of the buckets which is = 5 x d = **0.425 m**

d. **Depth of the buckets:**

Depth of the buckets which is = 1.2 x d = **0.102 m**

e. **No of buckets:**

(Z) = 15 + = **24**

---

12. The three Pelton turbine is required to generate 10,000 kW under a net head of 400 m. the blade angle at outlet is 15° and the reduction in the relative velocity while passing over the blade is 5%, If the overall efficiency of the wheel is 80 %, $C_v = 0.98$ and speed ratio = 0.46, then find: (i) the diameter of the jet, (ii) total flow in m^3/s and (iii) the force exerted by a jet on the buckets. If the jet ratio is not less than 10, find the speed of the wheel for a frequency of 50 hertz/sec and the corresponding wheel diameter.

Given:

No of jets = 3

S.P = 10000 kW

H = 400 m

= 15°

$V_{r2} = 0.95\ V_{r1}$

$_o = 0.80$

$C_V = 0.98$

$C_u = 0.46$

f = 50 hertz/sec

= 10

To Find:

d =?

Q =?

F_X =?

N =?

D =?

Solution:

a. Discharge:

$_o$ = 0.80

W.P = 8000 kW

Water power = , where,

W = g = 1000 x 9.81

$\mathbf{Q = 3.18\ m^3/s}$

b. **Diameter of jet:**

$V_1 = C_V = 87$ m/s

$Q = A\ V_1$

$Q = x\ d^2 \times V_1$

d = 0.125 m

c. **Force exerted by a jet on the wheel**

$u_1 = C_u = 40.75$ m/s

From inlet velocity triangle

$V_{r1} = V_1 - u_1 = 46.25$ m/s

$V_{w1} = V_1 = 87$ m/s

From outlet velocity triangle

$V_{r2} = 0.95V_{r1} = 44$ m/s

=

$V_{w2} = 1.75$ m/s

F_X =

= **94.075 KN**

d. **Wheel diameter and speed**

= 10

D = 1.25 m

$u = u_1 = u_2 =$

N = 620 r.p.m

Now using the relation

N = (60 x f) / p

p = 4.85

Take the next whole no, p = 5

N = (60 x f) / p

N = 600 r.p.m

$u = u_1 = u_2 =$

D = 1.3 m